

Gordon Surface

One B-spline face through a whole network of crossing curves -- every U curve and every V curve, interior ones included.



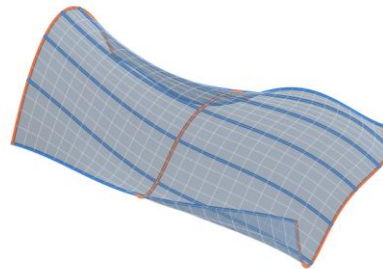
What it does

A loft passes through one family of sections; a boundary surface only through its edges. Gordon Surface passes through a full U x V network, so interior curves shape the face exactly where you drew them.

Use it when

- The shape is defined by curves in both directions
- Interior curves must lie on the surface
- One smooth face instead of patches

Gordon Surface 1 (Test Part Studio 1): one B-spline face through 5 U curves (blue) and 3 V curves (orange)



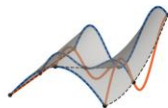
Gordon Surface 1: one face through 5 U and 3 V curves.



$$S = S_u + S_v - T$$

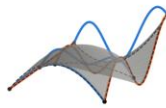
blue: U curves orange: V curves dashed: the surface along each curve dots: crossings

S_u: skin the U curves



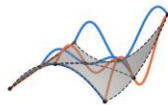
misses the U curves by 0.000, the V curves by 0.280

S_v: skin the V curves



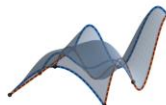
misses the U curves by 0.301, the V curves by 0.000

T: through the crossings only



misses the U curves by 0.301, the V curves by 0.280

Gordon: S = S_u + S_v - T



misses the U curves by 0.000, the V curves by 0.000

Two skins

S_u through the U curves (misses V); S_v through the V curves (misses U).

Minus the grid

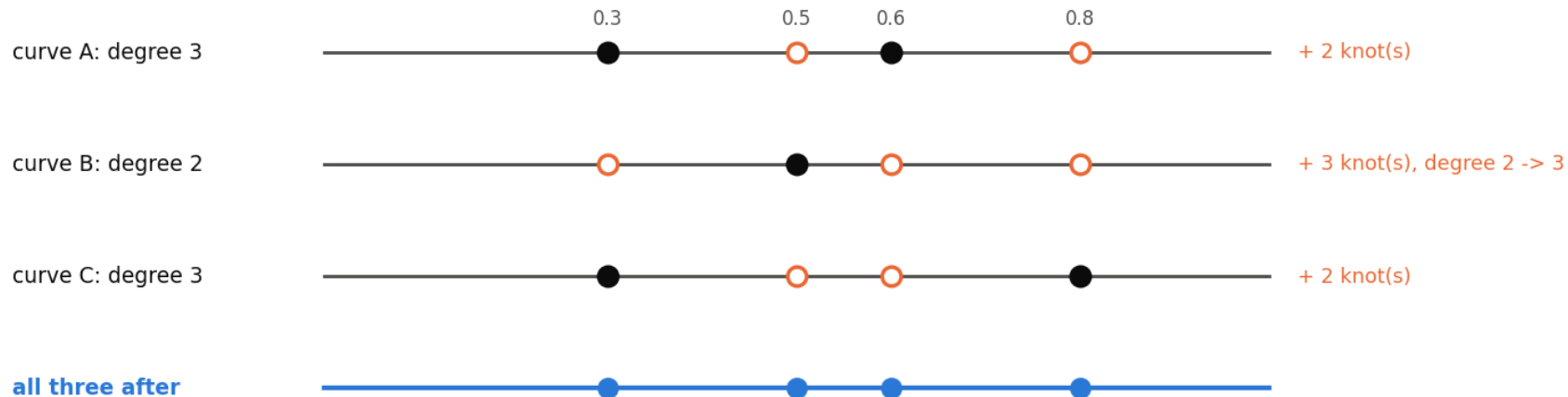
T passes through the crossings only; along each curve it cancels the other skin.

Exact on both

The sum reproduces every curve; done control point by control point once compatible.



Compatible curves



same degree, same knots, same control-point count -> corresponding control points can be blended; the curves' shapes do not change

Same degree

Every curve raised to the highest degree -
- exactly.

Same knots

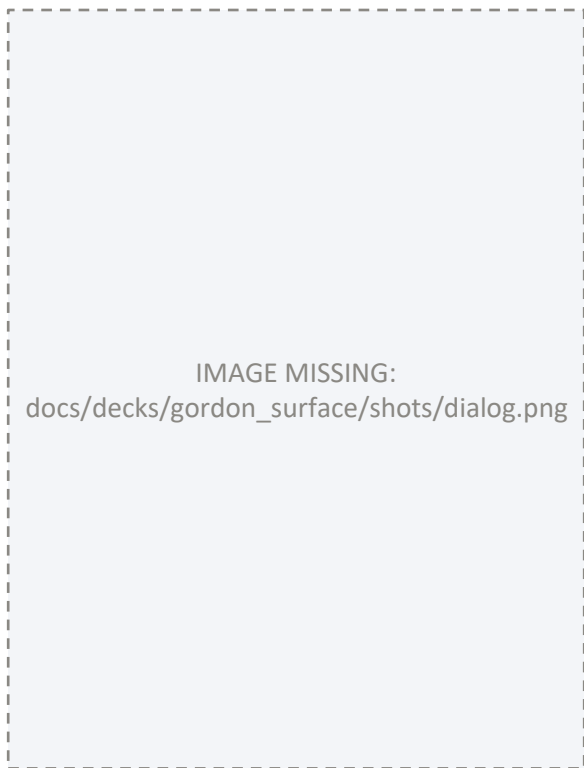
The union of all interior knots inserted
into every curve -- exactly.

Then blend

Corresponding control points can be
combined; the shapes do not change.



The dialog



#	Parameter	What it means
1	U-Curves	Up to 10 edges, in section order (the pick order is used).
2	V-Curves	The crossing family, in order.
3	Debug and details	Create S_u / S_v / tensor sheets, show curves (on), print, degree in u and v (3).



Outputs and messages

Outputs

Key	What it holds
Gordon surf	One sheet body.

Messages

	Message / what to do
Error	Need at least 2 curves Per family.
Error	(dimension mismatch) The network is inconsistent.



Tips and limits



Seed with Interior curves

Four boundaries -> interior curves that are guaranteed to cross.



Curves must cross

Every U must cross every V, in order; a near miss is used silently (midpoint of the closest points).



Averaged crossings

Uneven crossing parameters make the fit approximate; arcs' weights are ignored.



Where it lives

Document	<code>gordonSurface</code> (d0950ed72a894fbe35c27d43)
Tab	<code>gordonSurface</code> (+ <code>gordonCurveCompat</code>)
Tests	none yet
Examples	Test Part Studio 1: Gordon Surface 1, 2
Explainer	<code>docs/explainers/gordon_surface/gordon_surface_explained.md</code> , section 1.1, 1.2, 2.1
Related	Gordon Surface, Interior curves, Scaled Curve, Modify curve end, Pull surface, Simplify surface